Run the HAR LSTM/CNN script in Google Colab

Step-by-step walkthrough for running **har_lstm_cnn.py** on the UCI HAR dataset using a Google Colab notebook. Each numbered step is one thing to do; each gray block goes in its own Colab cell.

1. Open Colab and create a notebook

Go to **colab.research.google.com**, then File → New notebook.

2. (Optional) Switch to GPU

Runtime → Change runtime type → Hardware accelerator: **T4 GPU** → Save.

Not required — the script finishes on CPU in a few minutes — but the free GPU is faster.

3. Download the UCI HAR dataset

Paste in a new cell and run:

```
!wget -q "https://archive.ics.uci.edu/static/public/240/human+activity+recognition+using+smartphone
!unzip -q -o har.zip
!ls
```

If ls shows UCI HAR Dataset.zip (an inner zip), run this in a second cell:

```
!unzip -q -o "UCI HAR Dataset.zip"
!ls "UCI HAR Dataset"
```

You should see **train**, **test**, **features.txt**, etc. listed. If *ls* shows that, the data is in place.

4. Get the training script

Pick whichever option matches your situation:

(a) If the GitHub repo is public — clone it:

```
!git clone https://github.com/williamtran1889-ui/cs-ai.git
!cp cs-ai/har_lstm_cnn.py .
```

(b) If the repo is private — upload from your computer:

```
from google.colab import files
files.upload()   # click "Choose Files" and pick har_lstm_cnn.py
```

(c) Paste the script directly — open the file on GitHub, click "Raw", copy all the text, then in Colab do File → New file → name it **har_lstm_cnn.py** → paste → save.

After this step, run **!ls** in a cell. You should see both **har_lstm_cnn.py** and **UCI HAR Dataset/** in the same directory (*/content*).

5. Confirm dependencies

Colab has them preinstalled — this is just a sanity check.

```
import tensorflow, keras, sklearn, numpy
print("tf", tensorflow.__version__, "| keras", keras.__version__,
      "| sklearn", sklearn.__version__, "| numpy", numpy.__version__)
```

If this runs without error, you're good.

6. Run the script

```
!python har_lstm_cnn.py
```

Expect 30 epochs of LSTM training, then 30 epochs of CNN training, then two evaluation blocks (loss, accuracy, per-class F1, classification report, confusion matrix). On free T4 it's roughly 1-3 minutes total; CPU-only is about 5-10 minutes.

7. (Optional) Save the output

To capture the printed output to a file you can download:

```
!python har_lstm_cnn.py 2>&1 | tee results.txt
from google.colab import files
files.download("results.txt")
```

Common things that go wrong

FileNotFoundError: 'UCI HAR Dataset' → the dataset folder isn't in the same directory as the script. Run **!ls** and **!pwd** to check; if needed, **%cd /content**.

No module named 'flax' (only if you're running an old version of the script) → make sure you have the latest from the **claude/har-lstm-cnn-models-n8e7Q** branch — that import was removed.

Session timed out mid-training → Colab's free tier disconnects after ~90 min idle. Just rerun the cells; nothing persists between sessions anyway, so you'll redo steps 3-6.